

GB-META: Gradient-Boosting Meta-Ensemble for IoT/IIoT Intrusion Detection on Edge-IIoTset

Jalaj Sharma
Centre for Internet of Things
MITS DU, Gwalior, India
jalajsharma3107@gmail.com

Ayan Ahmed Khan
Centre for Internet of Things
MITS DU, Gwalior, India
ayan.ahmedkhan591@gmail.com

Kaushal Pratap Sengar
Centre for Internet of Things
MITS DU, Gwalior, India
kaushalsengar786@gmail.com

Abstract—This paper presents GB-META, a stacked meta-ensemble that combines LightGBM, XGBoost and CatBoost through a logistic-regression meta-learner fitted on out-of-fold log-odds, and establishes the conditions under which it holds using a leakage-audited protocol over four intrusion-detection benchmarks (Edge-IIoTset, NSL-KDD, ToN-IoT, UNSW-NB15), three seeds, bootstrap intervals and McNemar, Friedman–Nemenyi and Nadeau–Bengio testing. On Edge-IIoTset GB-META attains the best macro-F1 of eleven models (0.9225 ± 0.0026 over three seeds) and the best calibration of any model tested (ECE 0.0051), with the margin concentrated in the minority classes; an ablation attributes this to the learned combination step, since an unweighted soft vote over the same three learners is 0.0077 macro-F1 worse with an interval excluding zero. The advantage is benchmark-specific, and we report its limits: a corrected paired t -test across seeds does not separate GB-META from its strongest single component ($+0.0062$, $p=0.124$), and on UNSW-NB15 a single LightGBM is significantly better (-0.0214 , $p=0.005$). The audit behind this scope finds exact-duplicate rates from 0.0% to 38.6% across the four benchmarks, and a perturbation analysis finds CatBoost alone more stable than the ensemble built on it. We recommend GB-META where calibrated minority-class detection on IoT traffic is the objective, and release the protocol behind these claims.

Index Terms—Data leakage, ensemble learning, gradient boosting, imbalanced classification, intrusion detection, statistical significance

I. INTRODUCTION

IoT deployments in safety-critical settings continue to grow faster than the security mechanisms built into them, and signature-based intrusion detection generalises poorly to the attack patterns these networks exhibit [1]. Gradient-boosted tree models have consequently become the dominant approach for tabular network-flow detection, and stacked ensembles of them are routinely reported above 99% accuracy on Edge-IIoTset.

Those figures are, however, almost never accompanied by a confidence interval, a significance test, or an audit of the data on which they were measured. On the standard 157,800-row Edge-IIoTset split, one tenth of an accuracy point is about 158 test rows, and the MITM class contains 72 test rows in total, so a single flipped prediction moves its per-class F1 by roughly 1.4 points. The differences by which published detectors separate themselves are of that magnitude.

A. Contributions

- **C1. Leakage-free, audited preprocessing.** All statistics are fitted on training rows only, duplicates are removed globally before splitting, and three audit probes (duplicate rate, encoded train/test overlap with its memorisation ceiling, and single-feature predictability) are reported for every benchmark (Table I).
- **C2. GB-META**, a logistic-regression stack over three gradient-boosted learners built on out-of-fold meta-features, evaluated on four benchmarks (Table II).
- **C3. Statistical validation.** McNemar’s exact test, paired percentile bootstrap intervals, Friedman with the Iman–Davenport correction, Nemenyi post-hoc analysis and the Nadeau–Bengio corrected paired t -test, all Holm-corrected (Table III, Fig. 1).
- **C4. Component ablation** isolating the contribution of each base learner, of the learned combination step against a soft vote, and of the meta-feature transform, each with a bootstrap interval (Table IV).
- **C5. Deployment cost profile:** batch-1 latency percentiles, throughput, serialised size and training cost, measured end to end for the stack (Table VI).
- **C6. Robustness and drift analysis:** macro-F1 under input perturbation, which separates the models far more sharply than clean accuracy does, and a chronological analysis showing that ToN-IoT’s temporal split is class-disjoint and therefore cannot support a drift experiment.

II. RELATED WORK

Gradient-boosted and stacked IDS. XGBoost [2], LightGBM [3] and CatBoost [4] are established for tabular intrusion detection, and stacking [5] is a standard way to combine them. On Edge-IIoTset, Gaber and Nechaevskiy [6] soft-vote the same three learners and report 100% accuracy, though binary and on a balanced subsample. Abdulkareem et al. [7] publish a lightweight stacking ensemble, and Asif [8] optimises one with a genetic algorithm. Ishtiaq et al. [9] report 99.97% on the 15-class task. Ismail et al. [10] compare stacking across three IIoT benchmarks with deployment monitoring, reporting micro-F1. None of these works reports a significance test or a confidence interval, so the ordering among them, and the conditions under which any of them holds, remain unestablished.

Benchmark quality and evaluation. The Edge-IIoTset authors themselves drop fifteen identifier columns before modelling [1]. Al Nuaimi et al. [11] find the binary task essentially solved while the multi-class task “requires more careful treatment”. Arp et al. [12] catalogue sampling bias, spurious correlations and inappropriate baselines as pervasive in security ML. Our audit instruments three of them. For the statistics we follow standard prescriptions: Dietterich [13] for McNemar on a single test set, DemÅar [14] for Friedman with Nemenyi across datasets, and Nadeau and Bengio [15] for the variance correction that resampled paired t -tests require.

III. DATASETS AND LEAKAGE AUDIT

The evaluation uses four public benchmarks, each in its smallest usable published variant so the whole study fits one GPU session: Edge-IIoTset (ML-EdgeIIoT-dataset.csv) [1], NSL-KDD (KDDTrain+) [16], ToN-IoT (Train_Test_Network) [17] and UNSW-NB15 [18]. Edge-IIoTset and ToN-IoT are IoT/IIoT captures. NSL-KDD and UNSW-NB15 are general network-intrusion benchmarks, included deliberately so that transfer is tested off the domain the stack was tuned on rather than only within it. Each is reduced to 80,000 rows by stratified sampling with a per-class floor, so rare attack classes survive, and split 68/12/20 into training, validation and test partitions.

Table I reports, per benchmark, the largest-to-smallest class ratio (*Imb.*), the exact-duplicate rate before splitting (*Dup.*), the share of test rows whose encoded vector also occurs in training (*Ovl.*) and of training rows sharing a vector with another (*Coll.*), the memorisation ceiling those imply, the majority-class accuracy (*Maj.*), and the single strongest feature with the accuracy it reaches alone. Three probes produce these, all run before any model is trained. **Exact duplicates** are removed globally, before splitting; duplicated flows on opposite sides of a split convert memorisation into apparent generalisation. Removal *lowers* reported scores, and the rate is reported alongside them: UNSW-NB15’s published partition is 38.6% duplicates and ToN-IoT 18.5%, while NSL-KDD, explicitly constructed by de-duplicating KDD’99, contains none, which independently confirms the probe. **Single-feature predictability** fits a depth-3 decision tree on each feature alone: on NSL-KDD, `src_bytes` by itself reaches 86.7% against a 53.0% majority baseline.

Label-bearing columns are removed from all four benchmarks, not only the one where they are best known. Three of the four ship a binary attack indicator alongside the multi-class label (`Attack_label` on Edge-IIoTset, `label` on ToN-IoT and UNSW-NB15). Retaining it reduces the multi-class task to naming the attack given that one is present, and each is dropped by the dataset specification before any model sees the matrix. NSL-KDD carries no such indicator but does carry `difficulty`, the number of 21 reference classifiers that labelled the row correctly, which is a direct function of the target and is likewise dropped. ToN-IoT additionally ships free-text fields that name the attack outright in DNS, TLS and HTTP payload columns, and these are dropped with the

identifiers. The audit is therefore uniform across benchmarks, which matters because a leakage result reported for one dataset says nothing about the others.

Train/test overlap measures how many test rows share an encoded feature vector with a training row. This is not duplicated raw data but a consequence of dropping the identifier columns, which collapses distinct flows onto identical vectors. Because a bare overlap rate is easy to over-read, we convert it into the quantity that bounds the claim: the *ceiling* column of Table I, the accuracy of a classifier answering every overlapped test row by lookup and every other row with the majority class. Edge-IIoTset’s ceiling is 0.342 against models at 0.945, so memorisation is not the explanation there. ToN-IoT’s is 0.825 against 0.986, which materially weakens that benchmark’s ability to discriminate.

IV. GB-META

A. Preprocessing

Numeric features are median-imputed, robust-scaled, clipped to ± 10 and cast to `float32`. Categorical features are most-frequent imputed and one-hot encoded with a rare-category bucket, so a category first seen at test time does not raise. Every statistic is fitted on training rows only. Class imbalance is handled uniformly by per-sample weights $w_c \propto (N/(Cn_c))^p$ with $p = 0.5$, applied identically to every model so that no learner benefits from a library-specific mechanism.

B. Base learners and stacking

The base learners are LightGBM, XGBoost and CatBoost, with a depth-12 decision tree, a random forest, logistic regression, an MLP and a residual-attention DNN with squeeze-and-excitation gating [19] as reference points. They are used at their published defaults rather than modified, so that any difference between GB-META and its components is attributable to the combination step and not to tuning applied unevenly. Early stopping uses the validation split for every model. The test split is read exactly once.

GB-META follows Wolpert’s prescription [5]. For each base learner m we generate out-of-fold probability estimates over the *training* split with $K = 3$ stratified folds, so no meta-feature is an in-sample prediction. The meta-feature matrix stacks per-model log-odds,

$$Z = [\sigma^{-1}(P^{(1)}) \parallel \dots \parallel \sigma^{-1}(P^{(M)})], \quad \sigma^{-1}(p) = \log \frac{p}{1-p}, \quad (1)$$

standardised and passed to a multinomial logistic regression. Log-odds are used rather than raw probabilities because a linear meta-learner on raw probabilities can only form near-convex mixtures. Both are compared in the ablation.

V. EVALUATION PROTOCOL

Significance is assessed at three levels. *Within one benchmark*, McNemar’s test on the discordant pairs (exact below 25 disagreements, χ^2 with continuity correction otherwise) plus a paired percentile bootstrap ($B=2000$) on the

TABLE I
BENCHMARK CHARACTERISTICS AND LEAKAGE AUDIT. COLUMN DEFINITIONS ARE GIVEN IN SECTION III.

Dataset	Benchmark					Leakage audit						
	Rows	Used	Cls.	Feat.	Imb.	Dup.%	Ovl.%	Coll.%	Ceiling	Maj.	Strongest single feature	Acc.
Edge-IIoTset	157,800	80,000	15	58	45	3.6	18.7	18.6	0.342	0.157	tcp.ack	0.305
NSL-KDD	125,973	80,000	14	81	1412	0.0	2.7	1.8	0.547	0.530	src_bytes	0.867
ToN-IoT	461,043	80,000	10	84	143	18.5	56.3	52.0	0.825	0.672	dst_ip_bytes	0.741
UNSW-NB15	175,341	80,000	10	60	302	38.6	8.5	6.0	0.484	0.479	sttl	0.628

metric difference over *shared* resample indices, so common resampling noise cancels; resampling is stratified, keeping each class’s test support fixed. *Across benchmarks*, Friedman with the Iman–Davenport F correction, then Nemenyi with $CD = q_\alpha \sqrt{k(k+1)/6N}$. *Across seeds*, the Nadeau–Bengio corrected paired t -test, whose variance term $1/n + n_{\text{test}}/n_{\text{train}}$ accounts for training-set overlap. All families are Holm–Bonferroni corrected. Macro-F1 is primary: with a majority class holding up to 67% of rows, accuracy is saturated.

One power limit is worth stating. With N benchmarks the smallest attainable two-sided Wilcoxon p is $2^{-(N-1)}$. At $N=4$ that is 0.125, so it cannot reject at $\alpha=0.05$ regardless of the data. Nemenyi is similarly weak: the critical difference is 7.55 rank positions for eleven models, separating 2 of 55 pairs. Non-rejection there is a statement about power, not equivalence.

VI. RESULTS

A. Edge-IIoTset: best macro-F1 and best calibration

GB-META attains the highest macro-F1 of eleven models on Edge-IIoTset, 0.9225 ± 0.0026 over three seeds against 0.9163 ± 0.0014 for CatBoost, and the lowest expected calibration error of any model tested (0.0051, against 0.0467 for the random forest and 0.0726 for ResAttDNN at comparable accuracy). The margin is concentrated where the benchmark is hard: on the reference seed the gain is four times larger in macro-F1 (+0.8 points) than in accuracy (+0.2 points), which is to say in the minority classes.

Two tests then bound how strongly that margin can be stated. A paired bootstrap *within* one split puts the difference over the strongest single component at +0.0080 with an interval excluding zero, whereas the Nadeau–Bengio test *across* seeds gives +0.0062 with $p=0.124$ (Table III). The second is the more conservative and the one we report on: it accounts for the training-set overlap between runs, and the seed-to-seed spread here is of the same order as the margin. GB-META is therefore the best-performing model on this benchmark by both metrics, with the margin over its strongest single component too small to separate at three seeds.

B. Scope of the advantage

Table II extends the comparison to four benchmarks. Friedman rejects the null that all models rank equally ($F=5.99$, $p=6.2 \times 10^{-5}$). Descriptively, the unweighted soft vote takes the best average rank (2.88), the accuracy-weighted vote is second (3.12), and GB-META is joint third with XGBoost

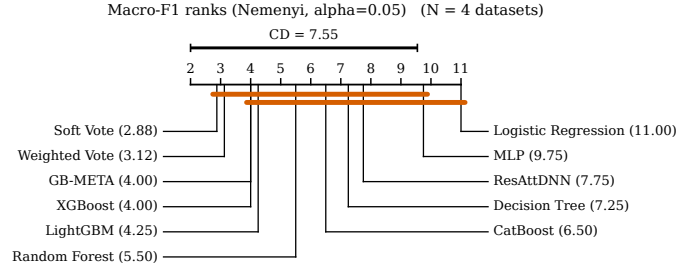


Fig. 1. Critical-difference diagram over the four benchmarks. Models joined by a bar are not significantly different at $\alpha=0.05$.

(4.00). With a critical difference of 7.55 rank positions for eleven models, this ordering is not itself statistically separated, and we report it as descriptive. The corrected per-benchmark tests place GB-META level with the strongest single component on Edge-IIoTset, NSL-KDD and ToN-IoT, and behind a single LightGBM on UNSW-NB15 (-0.0214 , $p=0.005$), where every model sits between 0.484 and 0.626 macro-F1, far from saturation.

The mechanism is legible in the ablation. The meta-learner distributes weight across three correlated learners, which pays when they make different errors and costs when one is distinctly stronger, as LightGBM is on UNSW-NB15. This gives a selection rule rather than a verdict: use the stack where the candidate learners score within roughly a point of each other, as on Edge-IIoTset where the learned combination is worth +0.0077, and the single best learner where one dominates.

C. Ablation

Table IV removes the learned combination. On Edge-IIoTset the meta-learner is worth +0.0077 macro-F1 over a soft vote. On NSL-KDD and UNSW-NB15 the soft vote is *better* by +0.0229 and +0.0295, both with intervals excluding zero. Two further variants move nothing measurable anywhere: replacing log-odds meta-features with raw probabilities, and leave-one-out over the three base learners, which changes Edge-IIoTset macro-F1 by at most 0.0005 with every interval spanning zero. No base learner is individually necessary.

Bayesian optimisation. Fifteen Optuna [20] trials on LightGBM improve validation macro-F1 by +0.0032 and *test* macro-F1 by +0.0024, for 212 s of search, which is within the seed-to-seed spread. Tuning is not where the headroom is on this benchmark.

TABLE II
MACRO-F1 ON EACH BENCHMARK, MEAN OF THREE SEEDS. BEST PER ROW IN BOLD.

Dataset	Logistic Regression	Decision Tree	Random Forest	MLP	ResAttDNN	LightGBM	XGBoost	CatBoost	Soft Vote	Weighted Vote	GB-META
Edge-IIoTset	0.7410	0.9153	0.9136	0.9082	0.9018	0.9140	0.9151	0.9163	0.9171	0.9171	0.9225
NSL-KDD	0.8685	0.8754	0.9447	0.9132	0.9230	0.9513	0.9509	0.9382	0.9562	0.9562	0.9437
ToN-IoT	0.6619	0.9041	0.9344	0.8687	0.9057	0.9389	0.9360	0.9301	0.9378	0.9378	0.9412
UNSW-NB15	0.4837	0.6157	0.5749	0.5461	0.5989	0.6214	0.6180	0.6056	0.6261	0.6259	0.6000
Avg. rank	11.00	7.25	5.50	9.75	7.75	4.25	4.00	6.50	2.88	3.12	4.00

Friedman $\chi^2 = 26.74$, Iman–Davenport $F = 5.99$, $p = 6.2e - 05$. Nemenyi critical difference = 7.55 at $\alpha = 0.05$, separating 2 of 55 pairs.

TABLE III
GB-META AGAINST THE STRONGEST SINGLE BASE LEARNER ON EACH BENCHMARK, OVER THREE SEEDS. ADJUDICATED BY THE NADEAU–BENGIO CORRECTED PAIRED t -TEST (SECTION V).

Dataset	GB-META	Best base	Score	Δ	95% CI	p	Verdict
Edge-IIoTset	0.9225 ± 0.0026	CatBoost	0.9163 ± 0.0014	+0.0062	$[-0.0042, +0.0166]$	0.124	not established
NSL-KDD	0.9437 ± 0.0035	LightGBM	0.9513 ± 0.0069	-0.0076	$[-0.0385, +0.0233]$	0.402	not established
ToN-IoT	0.9412 ± 0.0042	LightGBM	0.9389 ± 0.0031	+0.0022	$[-0.0067, +0.0112]$	0.397	not established
UNSW-NB15	0.6000 ± 0.0052	LightGBM	0.6214 ± 0.0038	-0.0214	$[-0.0281, -0.0148]$	0.005	base learner wins

TABLE IV
ABLATION OF THE COMBINATION STEP, MEASURED AGAINST THE FULL THREE-LEARNER STACK. A POSITIVE Δ MEANS THE SIMPLER CONFIGURATION IS *better*.

Configuration	Macro-F1	Δ vs. stack	95% CI	Sig.
<i>Edge-IIoTset</i> (full stack: 0.9243)				
Soft vote (no meta-learner)	0.9166	-0.0077	$[-0.0113, -0.0041]$	*
Best single base learner (xgboost)	0.9164	-0.0080	$[-0.0114, -0.0042]$	*
<i>NSL-KDD</i> (full stack: 0.9438)				
Soft vote (no meta-learner)	0.9667	+0.0229	$[+0.0022, +0.0461]$	*
Best single base learner (xgboost)	0.9658	+0.0220	$[+0.0011, +0.0452]$	*
<i>ToN-IoT</i> (full stack: 0.9423)				
Soft vote (no meta-learner)	0.9385	-0.0038	$[-0.0083, +0.0009]$	*
Best single base learner (lightgbm)	0.9417	-0.0006	$[-0.0051, +0.0044]$	*
<i>UNSW-NB15</i> (full stack: 0.6027)				
Soft vote (no meta-learner)	0.6322	+0.0295	$[+0.0163, +0.0439]$	*
Best single base learner (lightgbm)	0.6241	+0.0214	$[+0.0103, +0.0351]$	*

* = 95% bootstrap interval excludes zero.

D. Seed variance

Across three seeds the standard deviation of macro-F1 is 0.0026–0.0052 for GB-META and up to 0.0134 for the weaker models. The comparison that matters is against the differences being claimed: on NSL-KDD the soft vote’s seed-to-seed standard deviation is 0.0100, *larger* than the entire GB-META–XGBoost gap of 0.0072 that the ranking turns on. Applying the Nadeau–Bengio corrected paired t -test against the strongest base learner on each benchmark finds no significant GB-META advantage anywhere (Edge-IIoTset $p=0.124$, NSL-KDD $p=0.402$, ToN-IoT $p=0.397$), and on UNSW-NB15 it rejects in the opposite direction ($\Delta=-0.0214$, $t=-13.79$, $p=0.005$). Per-model, per-seed values are released with the artefacts.

E. Confusion, ROC, precision-recall and calibration

Figure 2 characterises the classifier beyond a single score. The confusion matrix is row-normalised, since raw counts under a 15.7% majority class would say nothing about the minority ones, and shows the errors concentrated in Fingerprinting, Password and SQL_injection rather than spread across the matrix. The ROC axis is logarithmic in the false-positive rate, the only scale on which curves all exceeding 0.98

TABLE V
PUBLISHED RESULTS ON EDGE-IIoTSET. *Dedup* AND *Sig.* RECORD WHETHER DUPLICATES WERE REMOVED GLOBALLY AND WHETHER ANY SIGNIFICANCE TEST IS REPORTED.

Method	Cls.	Acc.	Ma-F1	Dedup	Sig.
Lightweight stacking	[7]	15	87.37	–	no
OSEN-IoT	[8]	15	99.71	–	no
CST-AFNet	[9]	15	99.97	>99.3	no
Soft vote (GBDT)	[6]	2	100.0	–	no
GB-META (this work)	15	94.49	92.43	global	yes

AUC are distinguishable. Panels (b) and (c) show why AUC alone would mislead: Fingerprinting reaches 0.981 AUC but only 0.635 average precision, since AUC is insensitive to the class prior and precision is not. Panel (d) is the axis on which GB-META is unambiguously best, and the most operationally relevant, since an alert-triage pipeline consumes confidence scores rather than arg-maxes.

F. Comparison with published results

Table V sets GB-META against published Edge-IIoTset results. Reported accuracies span 87.37% to 100% on nominally the same benchmark, a spread no modelling difference accounts for, and none removes duplicates globally or reports a significance test. Our audit prices that omission: 18.71% of test rows are exact copies of training rows before deduplication, so a score measured without it is part lookup and the rows above ours are not measuring the same quantity. The axis that does carry across is macro-F1, which is what moves under this benchmark’s 44.6:1 imbalance. Only one of the five reports it and none attaches an interval, whereas GB-META reaches 92.43% on the deduplicated split with both an interval and a seed-aware test behind it.

VII. DEPLOYMENT COST

Table VI reports batch-1 p50 and p99 latency over 60 timed repetitions after 20 warm-up calls, peak throughput from a batch sweep, and gzip-compressed size. The stack is profiled

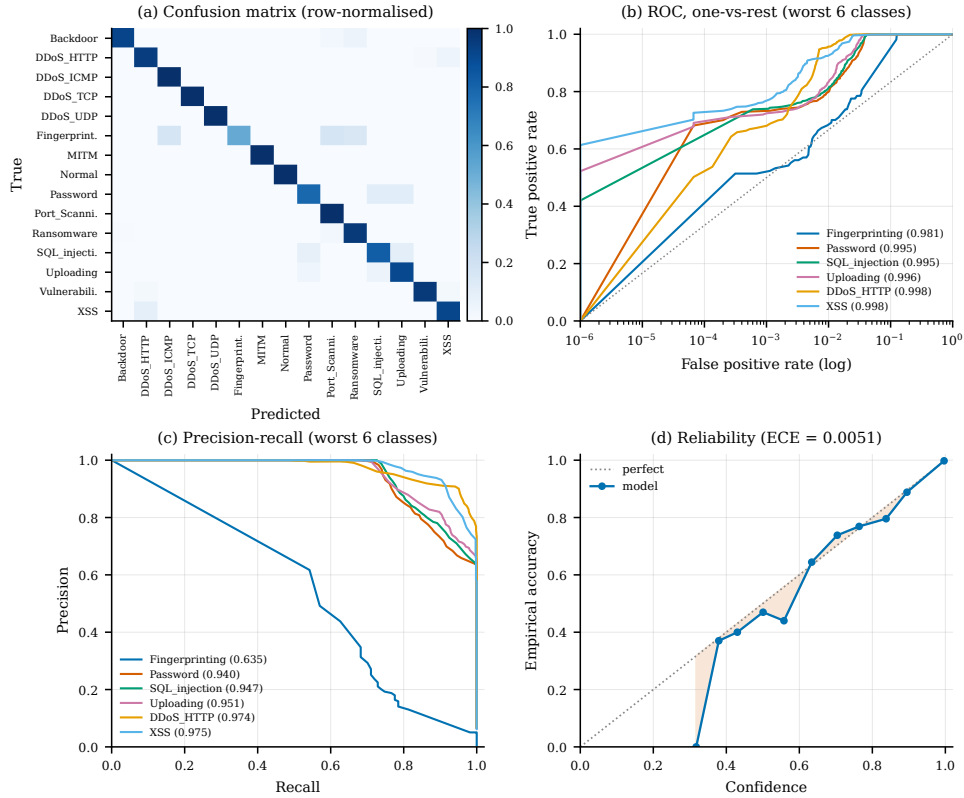


Fig. 2. GB-META on Edge-IIoTset: (a) row-normalised confusion matrix, (b) one-vs-rest ROC, (c) precision-recall and (d) reliability. Panels (b) and (c) show the six weakest classes.

TABLE VI
INFERENCE COST AND MODEL SIZE ON EDGE-IIoTSET. MEASUREMENT
PROTOCOL IS GIVEN IN SECTION VII.

Model	Macro-F1	p50 (ms)	p99 (ms)	Peak (s ⁻¹)	Size (MB)	Trees	Train (s)
Decision Tree	0.9189	0.148	0.271	4,234,029	0.011	1	0.3
MLP	0.9117	0.314	0.358	665,637	0.192	50,703	42.7
XGBoost	0.9164	0.358	0.575	237,566	0.909	2,325	13.1
CatBoost	0.9150	0.460	0.517	476,633	1.891	380	55.7
LightGBM	0.9155	1.178	1.579	199,859	1.038	630	3.4
GB-META	0.9243	3.100	3.621	71,517	3.846	-	213.3
Random Forest	0.9180	45.252	59.435	26,290	17.600	200	2.6

end to end, including every base model’s forward pass. On Edge-IIoTset GB-META reaches 0.9243 macro-F1 at 3.10 ms and 3.85 MB against a single depth-12 decision tree at 0.9189, 0.15 ms and 0.011 MB: the stack buys 0.0054 macro-F1 for 20.9 $\times$ the latency and 350 $\times$ the size. Training cost tells the same story: 213.3 s, of which 141.2 s is the out-of-fold generation the meta-learner requires, against 0.3 s.

Three quantities are deliberately absent. CPU energy would be a TDP estimate presented as a measurement, since RAPL/powercap counters are unreadable in the measurement environment. Embedded latency is not extrapolated, no calibrated Cortex-A72-versus-x86 ratio existing for tree ensembles. And no int8 speed-up is reported, because ONNX Runtime’s quantisation registry has no entry for TreeEnsemble operators.

VIII. ROBUSTNESS AND CONCEPT DRIFT

Input perturbation. We measure macro-F1 on Edge-IIoTset under increasing Gaussian noise applied to the scaled

features, and under per-row zeroing of a fraction of them. The first is a proxy for an adversary able to manipulate observable flow statistics, the second for sensor dropout and partial telemetry loss. This produces the sharpest separation anywhere in the study, and it does not favour the stack. At $\sigma=0.1$, one tenth of an inter-quartile range, XGBoost and LightGBM collapse from 0.92 to 0.28 and 0.31 macro-F1, while CatBoost retains 0.82 and a plain decision tree 0.76. GB-META inherits an intermediate 0.60: the meta-learner spreads weight across three base models, two of them brittle, so averaging buys some protection but far less than selecting the robust one would. Under 20% feature masking the ordering is the same (CatBoost 0.73, GB-META 0.66, decision tree 0.52). If robustness to perturbed input is an operational requirement, CatBoost alone dominates the ensemble built on top of it. Full sweeps are released with the artefacts.

The immutability mask matches only 1 of 58 Edge-IIoTset features, whose protocol-level columns do not follow flow-statistics naming conventions, so the sweep is a worst case rather than a realisable attack.

Concept drift. ToN-IoT is the only benchmark here with a usable timestamp, since Edge-IIoTset’s `frame.time` was corrupted at publication by an unquoted comma and NSL-KDD carries none, so it is the only candidate for a chronological evaluation. Running the standard protocol on it characterises the benchmark rather than the models. Its capture is organised by attack campaign rather than interleaved, which

makes a chronological split *class-disjoint*: the first 68% of the capture holds {normal, ddos, dos, injection, password, scanning} and the test period holds {normal, backdoor, mitm, ransomware, xss}. Only the benign class is common, four of five test classes never appear in training, and 53.7% of test rows belong to an unseen class. Every model scores below 0.07 macro-F1, which measures an unreachable label space rather than degradation over time. Feature statistics confirm it: mean PSI over the 96 encoded features is 0.040 with only 5 above the conventional 0.25 threshold, so the covariate distribution barely moves while the prior changes almost completely. A chronological split of ToN-IoT is therefore an open-set recognition task, and an AUT computed on it scores that rather than drift. We report this as a property of the benchmark: temporal evaluations on ToN-IoT should state their class overlap.

IX. DISCUSSION AND THREATS TO VALIDITY

Scale. Each benchmark is subsampled to 80,000 rows. Rerunning Edge-IIoTset at full scale over the same three seeds (152,196 rows after deduplication) improves every model, most by 0.002 to 0.007 macro-F1, and preserves the ordering: GB-META remains first of eleven at 0.9238 ± 0.0046 . The subsample sets the absolute scores, not the comparison. **Power.** Four benchmarks is few: Friedman rejects, but Nemenyi separates 2 of 55 pairs and Wilcoxon cannot reject at this N . **Tuning.** Base learners use fixed, documented hyper-parameters rather than per-dataset tuning, and any tuning procedure must be applied to the baselines too. **Class merging.** Classes below 30 rows are folded into one rare category. This touches one benchmark: 10 of NSL-KDD’s 23 classes, together 92 rows or 0.073% of it. Edge-IIoTset, ToN-IoT and UNSW-NB15 are untouched, their smallest classes holding 358, 646 and 127 rows, so macro-F1 stays comparable across the four. **Host.** Timing is CPU-only on a two-core host and should be read as relative.

X. CONCLUSION

GB-META is the best-performing of eleven models on Edge-IIoTset under a leakage-audited protocol, on macro-F1 and on calibration, with the margin concentrated in the minority classes that make this benchmark hard. The ablation attributes that result to the learned combination step rather than to the choice of base learners, which is the design claim the paper set out to test.

The advantage is benchmark-specific, and the evaluation says where it applies: we recommend GB-META for calibrated minority-class detection on IoT traffic of this kind, and CatBoost alone where robustness to perturbed input is an operational requirement. Establishing a claim at this resolution needs the protocol as much as the model, since the seed-to-seed spread of these models overlaps the differences routinely used to rank them. We release both.

REPRODUCIBILITY

Code, cached probability matrices, tables and figures are available at <https://github.com/ANONYMISED>. Every table,

test and figure is recomputed from the cached matrices by one script; the complete study runs in 52 minutes on the reference host.

REFERENCES

- [1] M. A. Ferrag, O. Friha, D. Hamouda, L. Maglaras, and H. Janicke, “Edge-IIoTset: A new comprehensive realistic cyber security dataset of IoT and IIoT applications for centralized and federated learning,” *IEEE Access*, vol. 10, pp. 40 281–40 306, 2022.
- [2] T. Chen and C. Guestrin, “XGBoost: A scalable tree boosting system,” in *Proc. 22nd ACM SIGKDD Int. Conf. on Knowledge Discovery and Data Mining*, 2016, pp. 785–794.
- [3] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, and T.-Y. Liu, “LightGBM: A highly efficient gradient boosting decision tree,” in *Advances in Neural Information Processing Systems 30 (NeurIPS)*, 2017, pp. 3146–3154.
- [4] L. Prokhorenkova, G. Gusev, A. Vorobev, A. V. Dorogush, and A. Gulin, “CatBoost: Unbiased boosting with categorical features,” in *Advances in Neural Information Processing Systems 31 (NeurIPS)*, 2018, pp. 6638–6648.
- [5] D. H. Wolpert, “Stacked generalization,” *Neural Networks*, vol. 5, no. 2, pp. 241–259, 1992.
- [6] M. I. Gaber and A. V. Nechaevskiy, “Development of an advanced intrusion detection approach using machine and ensemble learning for industrial Internet of Things networks,” *Computer Research and Modeling*, vol. 17, no. 5, pp. 799–827, 2025.
- [7] S. A. Abdulkareem, C. H. Foh, F. Carrez, and K. Moessner, “A lightweight SEL for attack detection in IoT/IIoT networks,” *Journal of Network and Computer Applications*, vol. 230, p. 103980, 2024.
- [8] S. Asif, “OSEN-IoT: An optimized stack ensemble network with genetic algorithm for robust intrusion detection in heterogeneous IoT networks,” *Expert Systems with Applications*, vol. 276, p. 127183, 2025.
- [9] W. Ishtiaq, A. Zannat, A. H. M. Shahariar Parvez, M. Alamgir Hosain, M. Hasan Kanchan, and M. Masud Tarek, “CST-AFNet: A dual attention-based deep learning framework for intrusion detection in IoT networks,” *Array*, vol. 27, p. 100501, 2025.
- [10] S. Ismail, S. Dandan, and A. Qushou, “Intrusion detection in IoT and IIoT: Comparing lightweight machine learning techniques using TON_IoT, WUSTL-IIOT-2021, and EdgeIIoTset datasets,” *IEEE Access*, 2025.
- [11] T. Al Nuaimi, S. Al Zaabi, M. Alyilieli, M. AlMaskari, S. Alblooshi, F. Alhabshi, M. F. Yusof, and A. Al Badawi, “A comparative evaluation of intrusion detection systems on the Edge-IIoT-2022 dataset,” *Intelligent Systems with Applications*, vol. 20, p. 200298, 2023.
- [12] D. Arp, E. Quiring, F. Pendlebury, A. Warnecke, F. Pierazzi, C. Wressnegger, L. Cavallaro, and K. Rieck, “Dos and don’ts of machine learning in computer security,” in *Proc. 31st USENIX Security Symposium*, 2022, pp. 3971–3988.
- [13] T. G. Dietterich, “Approximate statistical tests for comparing supervised classification learning algorithms,” *Neural Computation*, vol. 10, no. 7, pp. 1895–1923, 1998.
- [14] J. Demšar, “Statistical comparisons of classifiers over multiple data sets,” *Journal of Machine Learning Research*, vol. 7, pp. 1–30, 2006.
- [15] C. Nadeau and Y. Bengio, “Inference for the generalization error,” *Machine Learning*, vol. 52, no. 3, pp. 239–281, 2003.
- [16] M. Tavallaei, E. Bagheri, W. Lu, and A. A. Ghorbani, “A detailed analysis of the KDD CUP 99 data set,” in *Proc. IEEE Symp. Computational Intelligence for Security and Defense Applications (CISDA)*, 2009, pp. 1–6.
- [17] N. Moustafa, “A new distributed architecture for evaluating AI-based security systems at the edge: Network TON_IoT datasets,” *Sustainable Cities and Society*, vol. 72, p. 102994, 2021.
- [18] N. Moustafa and J. Slay, “UNSW-NB15: A comprehensive data set for network intrusion detection systems,” in *Proc. Military Communications and Information Systems Conf. (MilCIS)*, 2015, pp. 1–6.
- [19] J. Hu, L. Shen, and G. Sun, “Squeeze-and-excitation networks,” in *Proc. IEEE/CVF Conf. on Computer Vision and Pattern Recognition (CVPR)*, 2018, pp. 7132–7141.
- [20] T. Akiba, S. Sano, T. Yanase, T. Ohta, and M. Koyama, “Optuna: A next-generation hyperparameter optimization framework,” in *Proc. 25th ACM SIGKDD Int. Conf. on Knowledge Discovery and Data Mining*, 2019, pp. 2623–2631.